PHY201 Practice Exam on chapter(s): 7, 8

Name:____

These questions assess your understanding of the material from Chapters 7, 8 of OpenStax College Physics. Please write your name on the sheets of paper that you use. Carefully read each question. Credit is only awarded if you answer correctly and clearly show all major steps necessary to find your answer; credit is not awarded for missing work, unclear work, or the use of memorized equations absent from the equation sheet. Half credit may be awarded for insufficient work that suggests some degree of understanding. Half credit is awarded for correct numerical answers with incorrect or missing units.

1. Abigail, who has a mass of 63 kg, has jumped from a plane and is falling through the air at her terminal velocity, a speed of 56 m/s. How much work does the force of air resistance do against her over a distance of 4.5 m?

   ANSWER: __________________________

2. Abigail pushes on a 66.1-kg package on a roller-belt conveyor system with a constant force of 160 N. The package moves over a distance of 16.5 m and the opposing friction force averages 8.0 N during this motion. Calculate the net work done on the package.

   ANSWER: __________________________

3. Jeremy places a 0.13-kg toy car next to a spring on a track. He moves the car backward to compress the spring 9.2 cm, then releases the car, propelling it upward along a curved track. How fast is the car going after it is released, before it begins travelling up the slope? Since Jeremy greased the axles of the toy car, assume there is no friction. Jeremy's previous experiments indicate that the spring has a force constant of 160 N/m.

   ANSWER: __________________________

4. Scott, a professional heavyweight boxer, delivers a forceful punch with his fist and arm, combining for an effective punching mass of 2.53 kg. His fist strikes a heavy bag at a velocity of 7.8 m/s and comes to a halt 10.3 ms after initial contact. Determine the magnitude of the average net force that Scott exerts on the heavy bag.

   ANSWER: __________________________

5. John, a 94-kg ice-hockey goalie who is originally at rest, catches a 0.19-kg ice-hockey puck that is traveling at a velocity of 42.5 m/s. What is the net change in John's and the puck's kinetic energy during the collision? Assume friction is negligible.

   ANSWER: __________________________

6. Olivia works in a factory sorting packages on a roller-belt conveyor system. Feeling a bit impatient, she helps move a 13.0-kg package along by pushing it over a distance of 10.8 m with a constant force of 70 N. Before she touched the package, it was moving at a constant speed of 0.6 m/s. If the opposing friction force between the conveyor belt and the package averages 6.3 N, find how far the package travels after Olivia stops pushing it.

   ANSWER: __________________________

7. Sam is roller skating, skating over a 4.1-m-tall hill. Assuming that he is moving at a speed of 4.8 m/s at the top of the peak, what is his speed when he reaches level ground at 0 m? Assume that friction is negligible.

   ANSWER: __________________________

8. Ryan, a 84-kg ice-hockey goalie who is originally at rest, catches a 0.2-kg ice-hockey puck that is traveling at a velocity of 42.4 m/s. Calculate Ryan's speed after he catches the puck.

   ANSWER: __________________________

9. David lifts a 99-kg barbell from the ground to a height of 1.4 m. How much work was done on the barbell?

   ANSWER: __________________________

10. Nicholas pushes a lawn mower, exerting a constant force of 56 N at an angle of 20° below the horizontal. He pushes the lawn mower over a distance of 17 m on level ground. How much work does he do on the lawn mower over this distance?

ANSWER: ______________________________

11. Jeremy crashes his car into a tree at 18.7 m/s (about 41 mph) and comes to an abrupt stop in 0.26 s. Find the magnitude of the average force that the seatbelt exerts on Jeremy to bring him to a halt. Jeremy's mass is 76 kg, which is roughly equivalent to a weight of about 167 lbs.

ANSWER: ______________________________

12. A 40.3-kg block is resting on a so-called "heavy-duty" shelf at a height of 6.6 m above the ground. The shelf suddenly collapses and the block falls to the ground. What is the block's total mechanical energy at a height of 2.6 m, assuming negligible air resistance?

ANSWER: ______________________________

13. Train cars are coupled together by being bumped into one another. Suppose two loaded train cars are moving toward one another, the first having a mass of 143000 kg and a velocity of 0.42 m/s, and the second having a mass of 156000 kg and a velocity of -0.47 m/s, where the minus sign indicates the direction of motion. What is the velocity of the two train cars after they couple together?

ANSWER: ______________________________

14. Caleb pauses at the bottom of a 11.8-m-tall flight of stairs, then ascends it in 12.0 s, moving at the top at a speed of 3.7 m/s. What is Caleb's power output in this motion if his mass is 71 kg?

ANSWER: ______________________________

15. Isabella is riding on a roller coaster, ascending a steep track as she approaches the top of a 80.4-m peak. Assuming Isabella is at rest at the top of the peak, what is her speed when she reaches level ground at 0 m? Assume that friction is negligible.

ANSWER: ______________________________

16. Hannah runs her computer, which consumes energy at an average rate of 210 W, for 6.3 hours per day. In a month with 29 days, what must Hannah pay to cover the cost of the computer's electricity if the rate is 9.3 cents per kilowatt-hour? Report your answer in dollars.

ANSWER: ______________________________

17. Caleb is designing a skate park and wants to build a curved ramp (quarterpipe) that leads skaters a distance of 4.5 m to the top of a straight ramp. Caleb plans that skaters will drop into the quarterpipe from rest and launch off the straight ramp with a speed of 1.6 m/s. If the height of the quarterpipe is 2.5 m, what must the height of the straight ramp be? Experimental studies in skate science show that the average friction force per mass on skaters as they skate across this particular surface is 0.65 N/kg.

ANSWER: ______________________________

18. Mary is playing tennis when she throws a 50-g tennis ball up in the air, hitting it when it's at its peak to give it a speed of 45.2 m/s, 1.3° above the horizontal. What is the magnitude of the average force that Mary's raquet exerts on the tennis ball, given that the ball remained in contact with the racquet for 5.0 ms?

ANSWER: ______________________________

19. Two identical billiard balls strike a rigid wall with the same speed and bounce without any change of speed. The first ball strikes perpendicular to the wall. The second ball strikes the wall at an angle of 27.0° from the perpendicular, and bounces off at an angle of 27.0° from perpendicular to the wall. Calculate the ratio of the magnitudes of impulses on the two balls from their collision with the wall.

ANSWER: ______________________________

20. Joshua is driving an experimental car that can accelerate from 0 m/s to 16.8 m/s in 5.8 s. If the combined mass of Joshua and the car is 1920 kg, what is the average power that the engine outputs? Assume there are no drag forces.

ANSWER: ______________________________

21. Two billiard balls, specifically the ten-ball and the five-ball, move toward each other until they elastically collide. The five-ball begins at rest until a force of 36 N is applied to it for a brief 9 ms, at which point it is moving toward the ten-ball. Calculate the velocity of the ten-ball after it elastically collides with the five-ball, given that each ball has the same mass of 156 g and the velocity of the ten-ball is 0.7 m/s, moving toward the five-ball.

ANSWER: ______________________________

22. Adam steps off a platform onto the floor from a height of 2.4 m. If Adam has a mass of 87.9 kg and lands without bending his knees, his knee joints compress by 0.4 cm. Calculate the force acting on his knee joints. Note that since Adam steps off the platform, his initial y-velocity is 0 m/s. Considering that 1000 N is roughly equal to 225 lbs, do you think this hurts Adam's knees?)

ANSWER: ______________________________

23. Grace is riding on a roller coaster, ascending a steep track as she approaches the top of a 40.6-m peak. Assuming Grace is moving at a speed of 3.2 m/s at the top of the peak, what is her speed when she reaches level ground at 0 m? Assume that friction is negligible.

ANSWER: ______________________________

24. Sean is playing baseball and is sliding into second base. He has a mass of 83.3 kg and begins sliding at a speed of 8.5 m/s. The average friction force acting against him during his motion is 1190 N. What is the distance that he slides?

ANSWER: ______________________________

25. Ryan skateboards up a curved ramp (quarterpipe) and launches himself into the air. He begins his motion with a speed of 8.2 m/s at a total distance of 5.0 m away from the launch position. In order to launch himself from the top of the quarterpipe, Ryan does 390 J of work. Ryan has a mass of 65 kg and knows that the average friction force acting on him as he skates across this particular surface is 20 N. If the quarterpipe has a height of 0.3 m, how far above the quarterpipe does Ryan reach?

ANSWER: ______________________________

26. Lauren is a 72-kg athlete competing in a race. She begins from rest with an average acceleration of 1.9 m/s^2 for 3.7 s. What is her momentum at 3.7 s?

ANSWER: ______________________________

27. Eric is playing baseball and is sliding into second base. He has a mass of 95.0 kg and begins sliding at a speed of 5.5 m/s. Measurements show that he covers a distance of 5.2 m during his slide. What is the magnitude of the average friction force acting against Eric during his slide?

ANSWER: ______________________________

28. A 0.17-kg red box is slid eastward on a frictionless surface into a dark room, where it strikes an initially stationary green box, which has a mass of 0.21 kg. The red box emerges from the room at an angle of 51° north of east. The speed of the red box is originally 5.3 m/s and is 1.1 m/s after the collision. Calculate the angle at which the green box travels after the collision.

ANSWER: ______________________________

29. Matthew, a 69-kg ice-hockey goalie who is originally at rest, is hit by a 0.18-kg ice-hockey puck that is traveling at a velocity of 38.8 m/s. Matthew hits the puck in such a way that this may be considered an elastic collision. Calculate the speed of the goalie after he hits the puck.

ANSWER: ______________________________

30. A rocket with a mass at liftoff of 2.8 million kg burns fuel at a rate of 17300 kg/s. Its exhaust velocity was measured to be 3100 m/s. Calculate its acceleration at liftoff.

ANSWER: ______________________________

31. A 0.23-kg orange box is slid eastward on a frictionless surface into a dark room, where it strikes an initially stationary purple box, which has a mass of 0.32 kg. The orange box emerges from the room at an angle of 32□° north of east. The speed of the orange box is originally 6.6 m/s and is 2.0 m/s after the collision. Calculate the speed of the purple box after the collision.

ANSWER: ______________________________

32. John is designing a skate park and wants to build a curved ramp (quarterpipe) that leads skaters to a straight ramp. John plans that skaters will drop into the quarterpipe from rest and launch off the straight ramp with a speed of 4.6 m/s. If the height of the quarterpipe is 2.4 m, what must the height of the straight ramp be? Assume that the wheel bearings are well-made and lubricated, so friction can be neglected.

ANSWER: ______________________________

33. A 2.0-kg falcon catches a 1.7-kg dove from behind while they are flying. What is their velocity after impact if the falcon's velocity is initially 22.3 m/s and the dove's velocity is 8.1 m/s in the same direction?

ANSWER: ______________________________

34. Rachel normally requires 14000 kJ (about 3346 Calories) of food energy per day. If Rachel consumes 15400 kJ per day, she will steadily gain weight. How much time must Rachel spend bicycling per day, if she does not want to gain weight? Rachel's average power output while bicycling is 380 W. Report your answer in minutes.

ANSWER: ______________________________

35. Kevin places a 0.15-kg toy car next to a spring on a track. He moves the car backward to compress the spring 8.1 cm, then releases the car, propelling it upward along a curved track. How fast is the car going after it rises to a height of 0.145 cm above the starting position? Since Kevin greased the axles of the toy car, assume there is no friction. Kevin's previous experiments indicate that the spring has a force constant of 110 N/m.

ANSWER: ______________________________

36. Katherine works in a factory sorting packages on a roller-belt conveyor system. Feeling a bit impatient, she helps move a 11.9-kg package along by pushing it over a distance of 3.2 m with a constant force of 130 N. Before she touched the package, it was moving at a constant speed of 0.1 m/s. If the opposing friction force between the conveyor belt and the package averages 35.1 N, calculate the final velocity of the package.

ANSWER: ______________________________

37. Samantha drives a 1890-kg tractor that pulls a 520-kg log along the ground for 51 m. The rope connecting the tractor to the log makes an angle of 46□° with the ground and is acted on by a tensile force of 4500 N. How much work does Samantha's tractor perform during this motion?

ANSWER: ______________________________

38. Mary is playing tennis when she throws a 50-g tennis ball up in the air, hitting it when it's at its peak to give it a speed of 45.2 m/s, 1.3□° above the horizontal. What is the magnitude of the average force that Mary's raquet exerts on the tennis ball, given that the ball remained in contact with the racquet for 5.0 ms?

ANSWER: ______________________________